

STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	I-III Semester (2025-26)
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Course Outcome	CO5-External Sort Processing via Heap Sort Priority Structures

Case Study Topic:

IPL Ad-Impression Archive — External Data Progression Using Heap Sort Selection

Work Done Summary:

In this case study, we sorted large disk-resident files under limited memory conditions using a Heap Sort min-heap strategy. We traced the step-by-step extraction of minimum elements from sorted disk chunks, completed a working Java boilerplate using a priority heap, evaluated data block I/O pass operations for a fifty gigabyte workload under an eight gigabyte RAM constraint, and contrasted heap-based extraction passes against binary pairwise merge costs.

Implementation Details:

1. Created a stream tracking system to manage separate data array blocks using simple index cursors.
2. Built a custom data wrapper to connect each extracted numeric value back to its specific chunk identifier.
3. Used a priority queue min-heap to sort data elements automatically by value.
4. Populated the priority queue initially with the front element of every active block stream.
5. Implemented an extraction loop that prints the smallest value and replaces it with the next element from that same block.
6. Added logic validation checks to smoothly drop out exhausted data streams as they empty

Result: Screenshots/output

```
8 10 11 12 15 19 22 25 28 30 33 38 41 47 49 51 55 60 75 80 85 88 89 92  
  
=== Code Execution Successful ===
```

GitHub Link: https://github.com/Varnika1024/DSA_CASE-STUDY/blob/main/co5_assignment.java